

Forecasting Solar Power Generation Using Deep Learning Techniques: A Case Study from Stockholm, Sweden

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ABSTRACT

Accurate forecasting of solar power generation is essential for the reliable integration of photovoltaic (PV) systems into sustainable energy systems, particularly in regions with high weather variability. This study investigates short term solar photovoltaic (PV) power forecasting for Stockholm, Sweden, specifically at Stockholm Arlanda Airport, using machine learning (ML) techniques applied to long term reanalysis data. Ten years (2015–2024) of hourly MERRA 2 reanalysis data from Renewables.ninja provide meteorological inputs—ambient temperature, precipitation, snow variables, surface air density, cloud cover, surface irradiance, and top of atmosphere irradiance—and simulated DC PV output per square meter as the prediction target. Three models of increasing complexity are developed and compared: logistic regression, a shallow neural network with one hidden layer, and a deep neural network with two hidden layers. All models undergo systematic data cleaning, feature normalization, and hyperparameter tuning, and are evaluated using cross-validation. The study also critically discusses potential biases, uncertainties, and risks associated with AI-based energy forecasts and their implications for decision-making in sustainable energy systems. Results show that neural network based models significantly outperform logistic regression, with the shallow network offering the best trade off between accuracy and computational cost, while the deep network achieves the highest accuracy but exhibits seasonal bias in summer peaks. The work demonstrates how AI methods can enhance PV forecasting in Nordic conditions, while highlighting the importance of feature selection, data quality, and bias analysis for trustworthy power forecasting.

Keywords: solar power forecasting, machine learning, Photovoltaic systems.

1 Introduction

The rapid growth of solar photovoltaic (PV) applications for power generation has created a need for accurate forecasting of PV output to support secure and efficient operation of electricity grids. Solar output is highly variable and driven by complex interactions between solar irradiance, cloud cover, ambient temperature, and seasonal solar geometry, making forecasting a challenging but central problem in power system operation, electricity markets, and infrastructure planning [3, 7]. Traditional approaches based on physical models, statistical models, or numerical weather prediction (NWP) can provide useful baselines; however, they often fail to fully capture nonlinear and multivariate relationships under rapidly changing sky conditions. Against this backdrop, artificial intelligence (AI) and machine learning (ML) techniques have become popular tools for learning complex input-output relationships directly from data, enabling more accurate solar power forecasts across a wide range of climates and system configurations [14, 5, 2]. Machine learning techniques are well suited to solar power forecasting due to their ability to identify complex nonlinear relationships between weather variables, temporal patterns, and historical power output.

The objective of this work is to forecast solar power generation for Stockholm using machine learning

and deep learning techniques and to systematically compare the performance of models with different levels of complexity. Specifically, the study aims to:

1. Review relevant scientific literature to identify commonly used machine learning algorithms and input features for solar power forecasting.
2. Select appropriate input features based on the literature and data availability.
3. Collect historical solar power and meteorological data from *renewables.ninja* and the Swedish Meteorological and Hydrological Institute (SMHI).
4. Clean, preprocess, and prepare the dataset for modeling.
5. Develop and optimize three forecasting models:
 - (a) Logistic Regression,
 - (b) Shallow Neural Network (single hidden layer),
 - (c) Deep Neural Network (multiple hidden layers).
6. Tune model hyperparameters and apply cross-validation to ensure robust evaluation.
7. Compare the forecasting accuracy of the three approaches.
8. Discuss potential biases and inaccuracies in AI-based energy forecasts and propose mitigation strategies.

A wide range of machine learning algorithms have been applied to solar power forecasting, ranging from traditional regression models to advanced deep learning (DL) algorithms. Verma et al. [14] explored both regression-based and neural network-based approaches for solar power generation forecasting. Their study compared linear, logarithmic, and polynomial regression models with artificial neural networks (ANNs). The results showed that while regression models provided reasonable baseline performance, ANNs were better suited to capturing nonlinear dependencies between meteorological variables and PV output, leading to improved forecasting accuracy.

With the growing availability of time-series data and increased computational power, recurrent neural networks (RNNs) have become popular for solar forecasting tasks. Elsaraiti and Merabet [7] applied Long Short-Term Memory (LSTM) networks to solar power forecasting and compared their performance with a traditional multilayer perceptron (MLP). LSTMs were found to outperform MLPs due to their ability to model temporal dependencies and retain information from previous time steps, which is particularly important for capturing diurnal cycles and weather persistence effects.

Chang et al. [5] conducted a comprehensive comparison of multiple algorithms, including Linear Regression, Random Forest, LSTM, and Bidirectional LSTM (BiLSTM). Their results indicated that deep learning models, particularly BiLSTM, achieved superior accuracy by leveraging information from both past and future time steps in the input sequence. However, the authors also noted that simpler models such as Random Forests performed competitively in certain scenarios, highlighting the importance of model selection based on data characteristics and forecasting horizon.

In a Nordic context, Brester et al. [3] evaluated site-specific PV forecasting models using numerical weather prediction (NWP) data and local weather observations. Their work focused primarily on feedforward artificial neural networks (MLPs), with LSTM models evaluated in supplementary analyses. They also compared machine learning models with a physical baseline model. The study demonstrated that data-driven neural networks significantly outperformed physical models when high-quality NWP inputs were available, reinforcing the value of machine learning in regions with complex weather patterns such as Scandinavia.

Hybrid DL approaches have been explored as well. AlKandari and Ahmad [2] proposed an ensemble framework combining Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and autoencoder-based variants (auto-LSTM and auto-GRU). Their multi-layer stacked hybrid model (MLSHM) showed

improved robustness and accuracy compared to single-model approaches, particularly under highly variable weather conditions.

Based on the reviewed literature, several input features consistently appear as important for solar power forecasting. Meteorological variables are the most commonly used inputs. Verma et al. [14] included ambient temperature, wind speed, wind direction, precipitation, relative humidity, atmospheric pressure, and cloud cover. These variables influence PV efficiency and irradiance availability either directly or indirectly. Irradiance-related features are also essential; Verma et al. [14] explicitly used diffuse horizontal irradiance and direct normal irradiance, along with past PV output power. Chang et al. [5] also incorporated irradiance measurements and historical power data, demonstrating that including lagged PV output significantly improves forecast performance due to persistence effects.

Temporal features represent another important category of inputs. Time-based variables such as hour of day, day of year, and seasonal indicators were used by Verma et al. [14] to capture diurnal and seasonal patterns in solar generation. Based on the reviewed literature, the most effective feature sets typically combine meteorological variables, irradiance data, temporal indicators, and historical PV power output.

2 Materials and Methods

2.1 Case Study

The case study for this project is located in Stockholm, Sweden, with the reference site at Arlanda Airport (latitude: 59.6524° N, longitude: 17.9514° E). Stockholm was chosen due to the authors' familiarity with the location; however, it also represents a highly relevant environment for solar power forecasting because of its high-latitude position, strong seasonal variability, and frequent cloud cover.

The annual average global horizontal irradiation (GHI) in the Stockholm region is approximately 900–1,000 kWh/m², which is lower than values observed in southern Europe but still sufficient to support meaningful PV power generation. Solar production is highly concentrated in the summer months, during which PV systems can reach daily outputs of up to 6.18 kWh/day per kW of installed capacity. In contrast, winter production is minimal due to short daylight hours and snow coverage. This pronounced seasonal contrast introduces significant nonlinearities in the data. Arlanda Airport was selected as the reference site due to its open surroundings, minimal shading effects, and the availability of high-quality meteorological data.

2.2 Data Collection

Historical meteorological and solar power data were collected based on insights from the literature review and the requirements of machine learning model development. The primary data source used in this study is *renewables.ninja*, which provides modeled renewable energy output and meteorological variables derived from the MERRA-2 reanalysis dataset. A 10-year dataset covering the period from 2015 to 2024 was extracted for the Arlanda Airport location.

The *renewables.ninja* platform was selected because it is widely used in academic research, offers transparent and reproducible data generation methods, and ensures temporal consistency with minimal missing values. The use of reanalysis data enables long-term model training while maintaining physical coherence between meteorological variables and solar irradiance.

The output variable for the forecasting models is direct current (DC) electricity output, expressed in kWh/m². This variable represents the theoretical PV output before inverter losses and is commonly used in forecasting studies, making it suitable for model comparison and evaluation.

The selection of input features plays a critical role in determining forecasting accuracy. Input feature selection was based on previous studies and the climatic conditions of Stockholm. Ambient temperature (°C) was included because temperature affects PV module efficiency, with higher temperatures generally reducing electrical output. Total precipitation, including both rain and snow, influences atmospheric clarity and surface conditions that can reduce irradiance. Snow precipitation and snow mass accumulation may

partially or fully cover PV panels, significantly reducing power generation even during daylight hours in winter. Surface air density, represented through humidity-related parameters, affects solar radiation scattering and panel absorption; previous studies have shown that humidity-related variables improve forecasting accuracy, particularly under cloudy or moist conditions [14]. Cloud cover was also included as an input feature due to its strong influence on short-term variability in solar power production, as higher cloud cover directly reduces generated power.

Irradiance-related inputs were also considered, namely surface irradiance and top-of-atmosphere irradiance. Irradiance is one of the most critical inputs for solar power forecasting, as it directly determines PV output. All reviewed studies incorporate irradiance-related variables due to their direct link to solar generation. Surface irradiance directly governs PV power production, while top-of-atmosphere irradiance captures the theoretical maximum incoming solar radiation.

Wind-related variables, such as wind speed and wind direction, were excluded from the dataset. Although some studies include wind parameters [14], their impact on PV output is generally indirect and relatively weak compared to irradiance- and cloud-related variables. Wind primarily influences PV module temperature through convective cooling, an effect that is already partially captured by ambient temperature. Excluding wind-related features reduces model complexity and limits the risk of overfitting, particularly when training ML models.

2.3 Data Cleaning and Preparation

The dataset used in this project consists of 10 years of hourly historical data (2015–2024) for Stockholm, obtained from the MERRA-2 reanalysis dataset via *renewables.ninja*, with Arlanda Airport as the reference location. An hourly time step was selected to capture short-term variability in solar power generation while maintaining computational efficiency. Input variables include meteorological parameters (ambient temperature, precipitation, snow-related variables, air density, and cloud cover), irradiance data (surface irradiance and top-of-atmosphere irradiance), temporal features (year, day, and hour), and DC photovoltaic electricity output.

Data quality checks identified missing electricity output values during daytime hours, primarily within the first two months of 2015. These missing values were treated as missing at random and corrected using linear interpolation to preserve the physical relationship between irradiance and PV output, given known theoretical bounds and the near-linear response of PV systems to irradiance over short time intervals. No duplicate records were detected, and minor formatting inconsistencies were corrected. All features were subsequently normalized to a common scale to ensure numerical stability and balanced model training.

Table 1: Parameter Data Summary

Parameter	Min	Median	Maximum
Temperature (°C)	-18.318	6.757	30.7
RHOA	1.152	1.248	1.392
CLDTLT (Cloud Cover)	0	0.704	1
SWGDN (Shortwave Radiation)	0	4.037	849.19

Estimates of parameter’s normalized values post data correction:

Table 2: Normalized feature values before and after data correction

Before Data Correction				After Data Correction			
SWGDN	SWTDN	CLDTOT	Electricity	SWGDN	SWTDN	CLDTOT	Electricity
72.677	229.48	0.713	0	72.677	229.48	0.713	0.037
82.165	266.03	0.772	0	82.165	266.03	0.772	0.043
72.642	256.89	0.769	0	72.642	256.89	0.769	0.041
46.318	202.56	0.748	0	46.318	202.56	0.748	0.022

2.4 Forecasting Models

2.4.1 Logistic Regression

Logistic regression is a model that produce value between 0 and 1, It usually intended for classification task that predicts discrete outcome. But, it also capable to predict non-discrete outcome like PV power output. The equation used is a sigmoid function, which described in Figure 1.

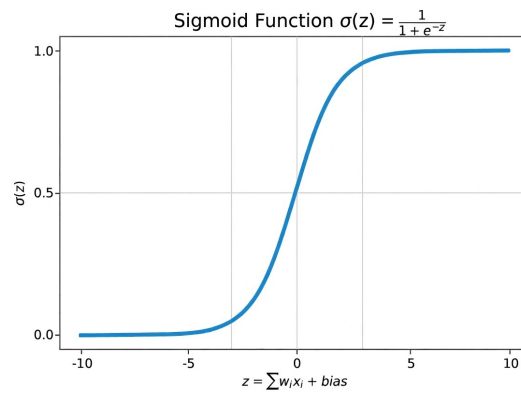


Figure 1: Sigmoid function used in logistic regression.

Predicted value of Y is calculated from normalized values of labeled data, and intital guess of W and b values. Then, the cost function is calculated. The W and b values will be updated to minimize the cost, the loop will continue until the maximum iteration. At the maximum iteration, the predicted value will be denormalized and stand as the prediction. The approach for applying this model is shown in Figure 2

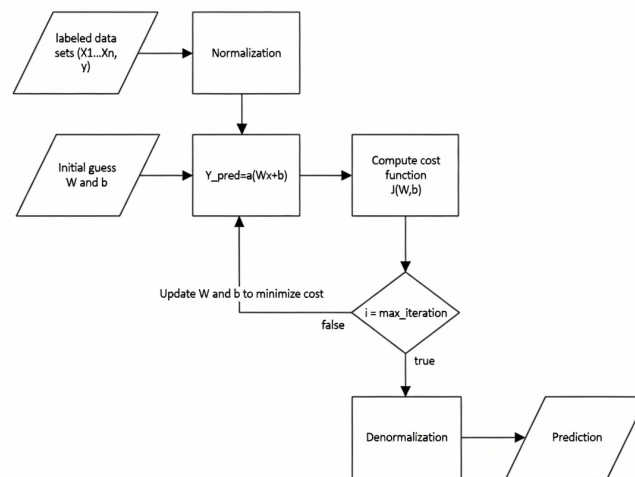


Figure 2: Logistic Regression Flowchart.

Predicted value of Y is calculated from normalized values of labeled data, and intital guess of W and b values. Then, the cost function is calculated. The W and b values will be updated to minimize the cost, the loop will continue until the maximum iteration. At the maximum iteration, the predicted value will be denormalized and stand as the prediction. The architecture of this model is shown in Figure 3

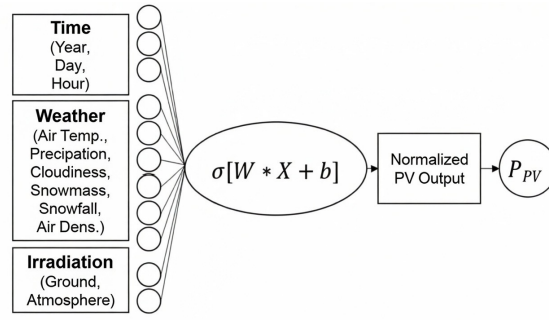


Figure 3: Logistic Regression Architecture.

2.4.2 Shallow Neural Network

The SNN model consists of one hidden layer and one output layer; the hidden layer employs the TanH activation function to model nonlinear atmospheric relationships with stable, zero-centered outputs, while the output layer employs the sigmoid function to produce bounded, probabilistic forecasts.

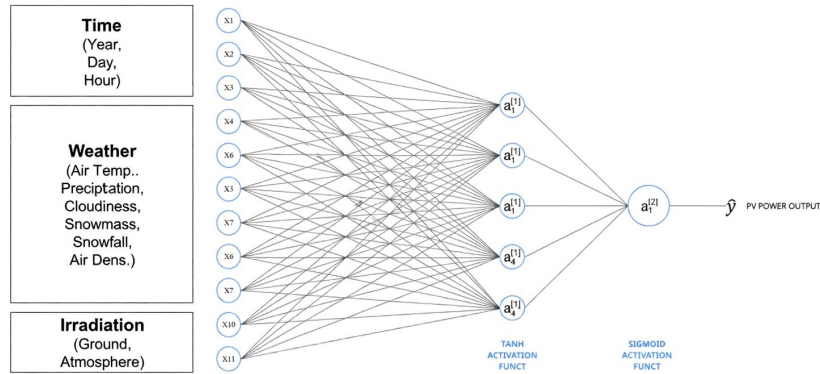


Figure 4: SNN Architecture.

2.4.3 Deep Neural Network

A DNN expands the beforementioned SNN with one or more layers. For the DNN in this project for only two layers were chosen, as a very good accuracy was already achieved with only a Shallow NN and by increasing the number of layers, the accuracy improves only slightly, but the computational time rises significantly. Similarly to the SNN 11 Input parameters were used, a non-linear activation function (ReLU) for the first two layers and a sigmoid output function for the last layer.

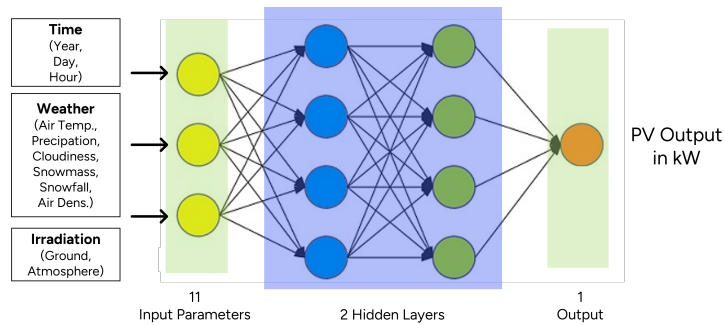


Figure 5: DNN Architecture.

3 Results

3.1 Logistic Regression

The logistic regression model is tested under different learning rate, Figure 6 shows the accuracy and solving time for various learning rates.

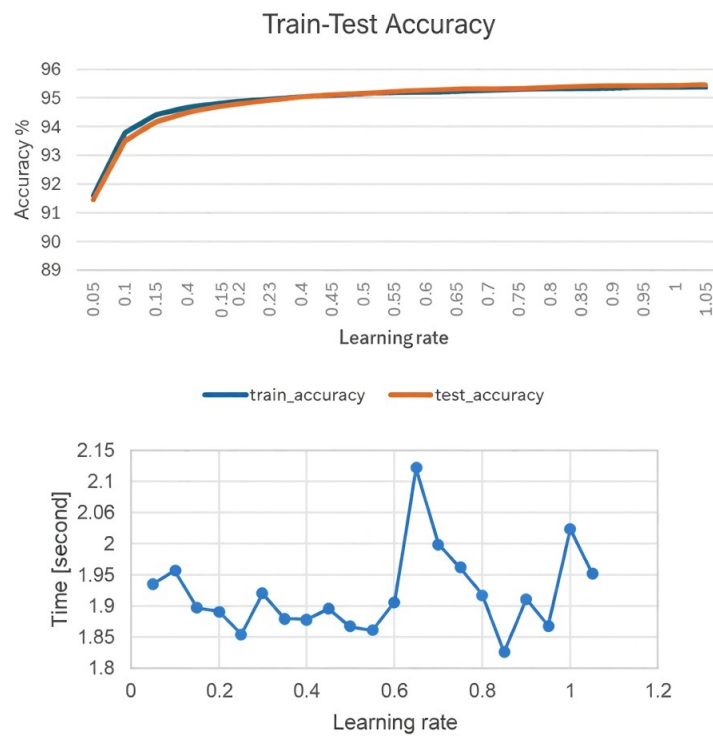


Figure 6: LR Learning Rate.

Learning rate 0.85 is picked as optimum value since the accuracy is high while requiring the least computation time. Learning curve from the value is shown in Figure 7

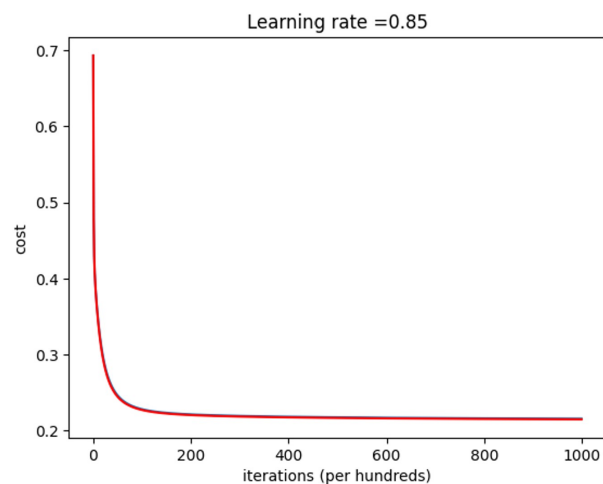


Figure 7: LR Learning Curve.

The model achieve good fit, since there is no indication of overfit nor underfit from the learning curve. By

1000 iterations, the model has 95.34% of accuracy and 88.6% R^2 . Indicated model is good and able to capture the variance of the data. Figure 8 shows residual plot of the model. In summer time, the model tends to overestimate the PV output while in winter time the model underestimate the value.

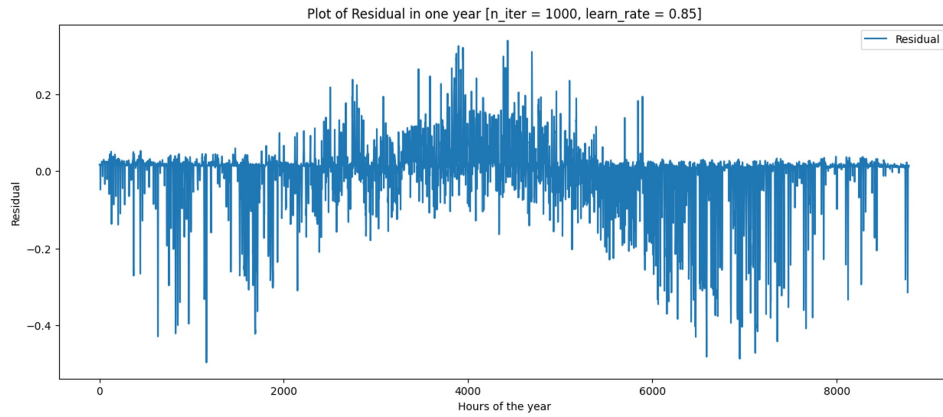


Figure 8: LR Residual Plot.

3.2 Shallow Neural Network

Figure 9 shows that tuning the learning rate (0.01–1.0) for a fixed 11–6–1 architecture identifies 0.85 as optimal with stable computation time, while tuning the hidden units at this learning rate confirms strong performance around 6 units but with nearly linear increases in computation time, highlighting the accuracy–complexity trade-off.

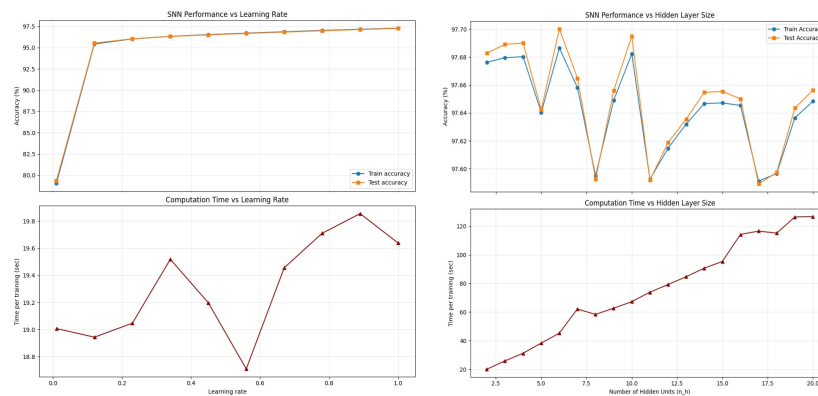


Figure 9: SNN Tuning.

Learning rate 0.85 is picked as optimum value since the accuracy is high while requiring the least computation time. Learning curve from the value is shown in Figure 10.

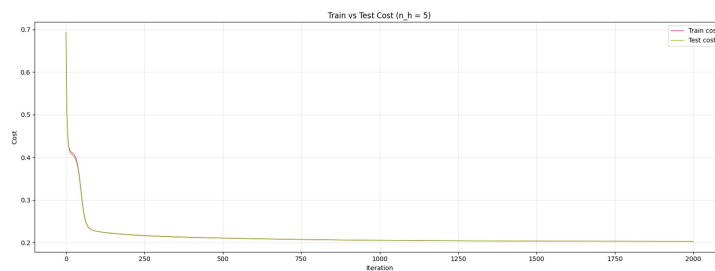


Figure 10: SNN Learning Curve.

The model demonstrates a good fit, as the learning curve shows no signs of overfitting or underfitting; after 1000 iterations, it achieves an accuracy of 97.09% and an R^2 of 94.21%, indicating strong capability in capturing data variance. Figure 11 presents the residuals, revealing a seasonal bias where PV output is slightly overestimated in summer and underestimated in winter.

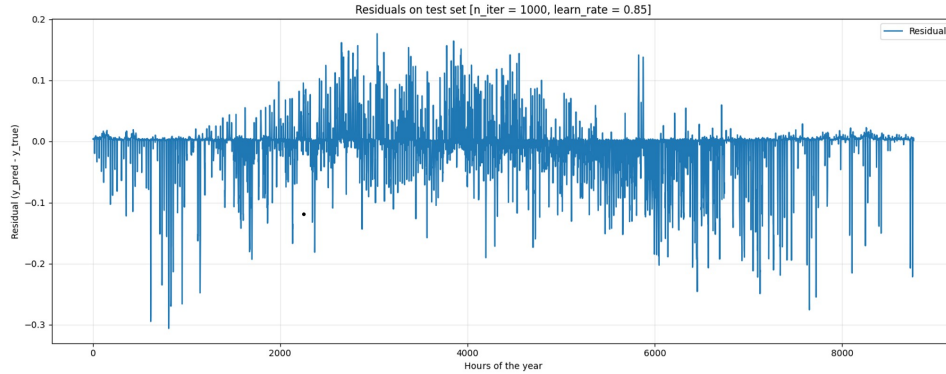


Figure 11: SNN Residual Plot.

The SNN with 6 hidden units shows consistently high and stable performance across all folds, achieving an average test accuracy of 97.35% with very low variance, indicating good generalization.

Table 3: 5-Fold Cross-Validation Results for SNN with 6 Hidden Units

Fold	Train Accuracy (%)	Test Accuracy (%)
1	97.35	97.37
2	97.35	97.37
3	97.36	97.33
4	97.36	97.35
5	97.36	97.34
Average	97.36 ± 0.01	97.35 ± 0.02

3.3 Deep Neural Network

To determine the best DNN structure, two different hyperparameters were tuned while using a 5-fold CV: the learning rate and the number of units per layer.

To reduce the hyperparameter tuning time, only 50 iterations were used, and the results were applied to the final model with 1000 iterations.

As already mentioned, increasing the number of iterations even further would improve the accuracy slightly, but would drastically increase the computation time. By using Pareto optimization with grid search, an optimal point was estimated, where the computation time is not too high but the accuracy improvement is significant. This point resulted to be with hidden layer sizes of 20 and 10 units, for the first and second layer each.

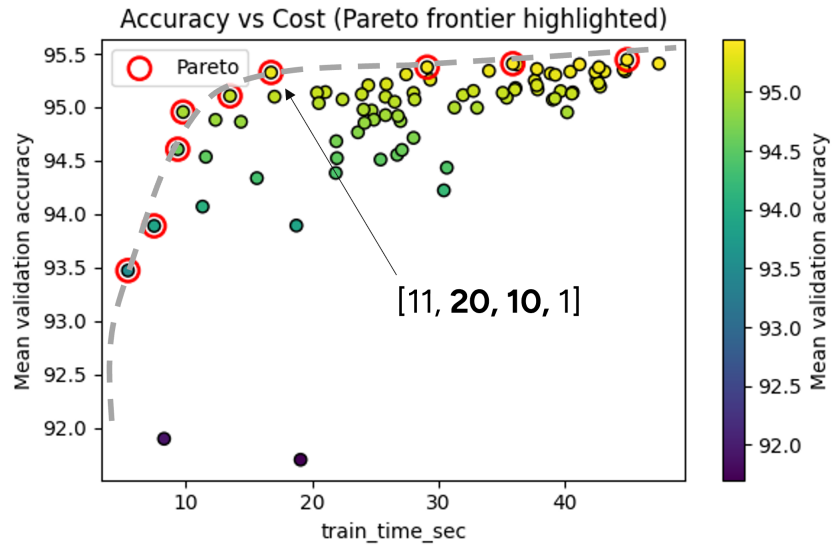


Figure 12: Pareto optimization to determine the best number of units per layer

By optimizing the learning rate, similar to the other models, a drastic increase between 0.1 and 0.4 can be seen, but the accuracy improves only slightly after. According to the graph, a learning rate of 0.85 was chosen, making it the last remaining parameter, for the final DNN model.

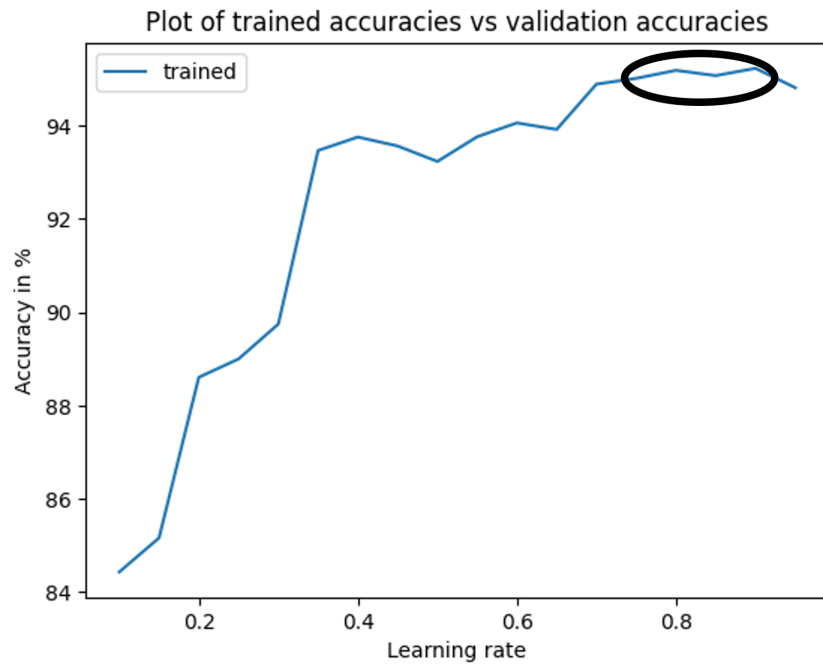


Figure 13: Optimization of the learning rate with 50 iterations and 5 Folds

When doing hyperparameter tuning, the residual plot of the test year was also analyzed and it can be seen, that a slight overestimation in summer and underestimation in winter is still happening, but time dependency was managed to be reduced, when comparing the DNN model it to the SNN and the LR.

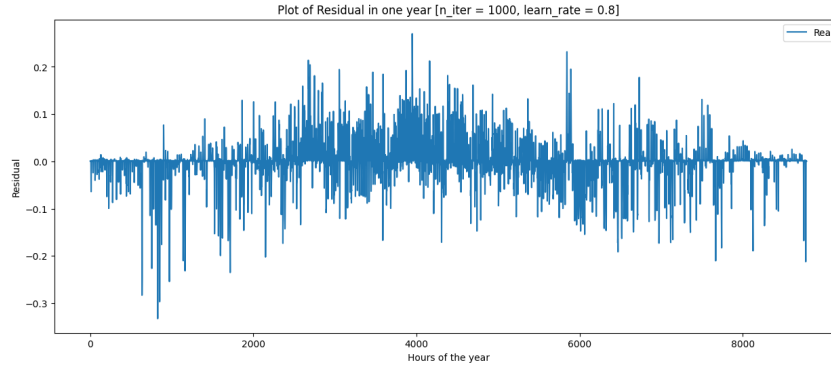


Figure 14: Residual Plot over one tested year with a clear summer and winter tendency

3.4 Results Comparison

The LR model results in the lowest accuracy and R^2 score; however, it delivers results in only 6 seconds while exhibiting a clear trend in the residual analysis. The SNN performs best overall, achieving very high accuracy and an R^2 score exceeding 94%, while requiring only half the training time of the DNN. The highest accuracy of 97.81% is achieved by the DNN, accompanied by an R^2 score of 96.8%; however, this performance comes at the cost of the longest computation time.

Table 4: Comparison of Forecasting Models Performance

Metric	LR	SNN	DNN
Accuracy (%)	95.34	+1.75	+2.51
R^2 Score (%)	88.6	+5.62	+8.19
RMSE	0.064	-0.019 (-29%)	-0.03 (-47%)
Computation Time (s)	5	52	102

4 Discussion

The results obtained from this study demonstrate that machine learning (ML)-based models are highly effective for short-term solar power generation forecasting and remain consistent under Nordic climatic conditions. Traditional regression-based approaches, such as logistic regression (LR), demonstrated comparatively lower predictive accuracy due to their limited ability to capture non-linear relationships between meteorological variables and PV power output. Similar conclusions have been drawn in previous studies, where regression models were primarily recommended as baseline or benchmark methods rather than high-accuracy forecasting tools [1, 15].

Both the shallow neural network (SNN) and deep neural network (DNN) achieved high accuracy, with R^2 values exceeding 94% and 96%, respectively, consistent with findings reported across multiple comparative studies [5, 12]. Prior research has consistently shown that neural networks outperform regression-based models when weather and irradiance inputs are incorporated, particularly for site-specific PV forecasting applications [3, 14]. The accuracy levels achieved in this study, which are approximately 97%, are comparable to those reported in recent deep learning-based forecasting studies conducted across different climatic regions [7, 8, 10].

Despite the strong overall performance, the DNN exhibited a systematic tendency to underestimate summer peak generation. This limitation has been widely documented in the literature and is often attributed to smoothing effects inherent in deep architectures, uncertainties in irradiance estimations, and limitations associated with numerical weather prediction (NWP) data [3, 6, 11]. Hybrid and ensemble approaches have been shown to mitigate such biases by combining complementary model strengths and improving sensitivity to extreme conditions [2, 4].

Several potential sources of uncertainty were identified, including missing data imputation using linear interpolation, exclusion of wind-related parameters, and reliance on reanalysis-based weather datasets [13]. While linear interpolation preserved physical consistency between cloud cover and irradiance, it may have reduced short-term variability, which affects peak prediction. These findings highlight the importance of rigorous data quality control, continuous model monitoring, and hybrid or ensemble approaches to mitigate bias and improve robustness in operational forecasting systems [6, 7].

From a system-level perspective, forecasting bias poses important challenges for sustainable energy systems. Underestimating PV generation can result in conservative grid operation, higher reserve requirements, and increased costs, whereas overestimating generation may threaten grid reliability and system adequacy [9, 14]. These issues highlight the need for robust model validation, continuous performance monitoring, and bias-aware forecasting when AI-based models are used for energy planning and operational decision-making.

5 Conclusion

This study evaluated multiple ML models for solar power generation forecasting using long-term meteorological and irradiance data for Stockholm, Sweden. The results demonstrate that neural network-based models significantly outperform traditional regression approaches in capturing the non-linear and seasonal characteristics of PV power generation. While LR provided fast computation and reasonable baseline performance, both the SNN and DNN achieved high predictive accuracy across all seasons.

Among the evaluated models, the SNN emerged as the most balanced solution, offering near deep-learning accuracy with substantially lower computational cost. Although the DNN achieved the highest overall accuracy, it exhibited systematic underestimation of summer peak generation, highlighting the importance of addressing seasonal bias in AI-based forecasting models. Overall, the models provided robust estimates of annual PV production, supporting their applicability for energy planning and operational analysis.

The findings of this work underscore the importance of accurate and reliable solar power forecasting for sustainable energy systems, particularly as AI-based models increasingly inform grid operation, storage sizing, and energy planning. Future work should focus on hybrid and ensemble approaches, improved irradiance representation, and feature ablation studies to better understand model sensitivity and reduce systematic bias. Strengthening data quality control and bias mitigation strategies will be essential to ensure trustworthy AI-driven decision support in the transition toward resilient and sustainable energy systems.

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